

TBILISI, Georgia

WMO: 375490

Lat: 41.7500N

Lon: 44.7694E

Elev: 1401

StdP: 13.97

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.5	26.4	14.3	11.9	30.0	17.4	13.9	32.5	43.8	40.2	38.7	40.6	6.4	320	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.0	95.3	73.0	92.9	72.3	90.1	71.6	76.5	90.5	74.8	88.5	73.1	86.3	10.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	126.3	85.3	70.4	117.7	83.1	68.7	111.2	81.3	41.1	90.7	39.4	88.8	37.7	86.1	92.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
43.4	36.2	31.3	19.2	99.7	4.0	2.5	16.4	101.5	14.0	103.0	11.8	104.4	8.9	106.3		
			18.0	79.6	3.8	4.0	15.2	82.5	13.0	84.8	10.9	87.1	8.1	90.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.9	38.1	40.6	46.7	55.4	64.9	73.3	78.3	78.8	70.2	59.7	47.7	40.2
	DBStd	15.66	4.99	5.97	6.15	6.86	5.74	5.36	4.59	5.14	5.73	6.21	6.23	5.93
	HDD50	1227	370	266	138	24	1	0	0	0	0	6	114	308
	HDD65	3949	835	683	566	296	72	6	0	0	19	182	520	770
	CDD50	4122	0	3	37	186	464	700	879	894	605	307	44	3
	CDD65	1368	0	0	0	7	69	256	414	429	174	19	0	0
	CDH74	12460	0	0	3	98	507	2179	4031	4304	1228	109	1	0
	CDH80	5314	0	0	0	16	103	852	1864	2078	390	11	0	0
(14) Wind	WSAvg	11.7	10.1	13.0	13.1	12.1	11.9	14.3	12.4	12.0	11.1	10.4	9.8	10.4
(15) Precipitation	PrecAvg	23.0	0.9	1.0	1.5	2.8	3.4	3.4	1.8	1.7	1.7	2.1	1.5	1.0
	PrecMax	31.5	2.1	2.2	3.3	4.8	8.0	6.7	3.4	3.8	4.5	5.7	3.6	3.6
	PrecMin	18.6	0.1	0.3	0.2	1.0	1.2	0.6	0.2	0.2	0.1	0.4	0.1	0.1
	PrecStd	3.6	0.5	0.5	0.8	1.1	1.3	1.5	0.8	1.0	1.1	1.2	0.9	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	62.7	71.6	82.0	87.0	95.0	98.2	99.4	91.9	81.0	68.4	59.9
		MCWB	46.5	50.1	56.9	63.2	68.6	71.8	74.9	74.0	71.6	65.3	55.9	49.0
	2%	DB	52.0	57.9	65.8	76.6	82.6	91.0	95.0	96.0	88.0	77.0	64.4	54.5
		MCWB	44.0	47.6	53.2	61.3	66.0	70.6	73.2	73.3	69.8	63.3	54.0	46.4
	5%	DB	48.6	53.8	61.1	72.0	79.6	88.1	92.5	93.5	84.4	73.4	60.8	51.2
		MCWB	42.0	45.4	50.4	58.9	65.2	69.9	72.7	72.3	68.1	62.1	52.3	44.5
	10%	DB	46.1	50.1	57.4	67.9	76.3	85.2	89.6	90.9	81.2	69.9	57.2	48.4
		MCWB	40.2	43.0	48.4	56.6	63.7	69.1	72.1	71.9	67.0	60.3	50.7	43.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.0	52.7	58.2	65.2	71.7	75.6	78.5	79.6	75.3	68.0	58.2	50.6
		MCDB	55.1	60.5	69.3	78.5	82.5	89.1	93.6	93.2	87.7	77.5	66.0	57.1
	2%	WB	44.8	48.9	54.1	62.4	68.0	73.0	76.3	77.1	72.0	65.2	55.5	47.6
		MCDB	50.6	56.1	63.4	74.0	78.9	87.0	91.2	90.7	83.7	73.9	61.2	52.6
	5%	WB	42.6	45.9	51.5	59.8	66.1	71.4	74.7	75.1	70.2	63.3	53.4	45.4
		MCDB	47.5	52.5	59.8	70.3	77.1	84.4	88.9	88.4	81.3	70.8	58.4	49.7
	10%	WB	40.8	43.5	49.3	57.5	64.5	70.0	73.0	73.3	68.2	61.6	51.5	43.8
		MCDB	45.1	49.1	55.9	66.1	74.7	82.5	86.2	86.6	78.0	68.0	56.3	47.6
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	14.7	16.7	18.0	18.3	19.7	19.4	20.0	18.5	16.1	15.2	12.9
		MCDBR	18.3	22.4	24.6	25.7	23.8	24.3	24.0	24.7	23.6	21.8	21.4	17.5
		MCWBR	12.4	14.0	14.6	13.6	11.8	9.9	9.8	9.9	10.1	10.9	12.7	11.1
	5% WB	MCDBR	17.1	20.5	21.9	23.7	21.4	22.1	21.7	22.7	21.4	19.8	17.8	15.0
		MCWBR	12.2	13.6	13.8	13.4	11.5	10.0	10.1	10.6	10.2	10.8	11.5	10.0
(40) Clear-Sky Solar Irradiance	taub		0.332	0.369	0.416	0.481	0.449	0.439	0.481	0.476	0.424	0.402	0.352	0.330
	taud		2.319	2.204	2.115	1.978	2.145	2.232	2.130	2.136	2.262	2.259	2.354	2.344
	Ebn at Noon		254	260	260	251	264	266	253	250	256	245	244	244
	Edh at Noon		28	36	45	55	48	44	48	46	38	34	27	25
(44) All-Sky Solar Radiation	RadAvg		592	879	1102	1369	1657	1933	1892	1742	1364	910	642	510
	RadStd		68	122	120	185	126	138	162	182	119	102	75	65

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+2.86	N/A	+2.14	+2.61	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page